GEOG 172

Reproducing the Openshaw Effect

Let's reproduce Openshaw's Effect in Rstudio and examine the distribution of the correlation.

Lecture 5-2



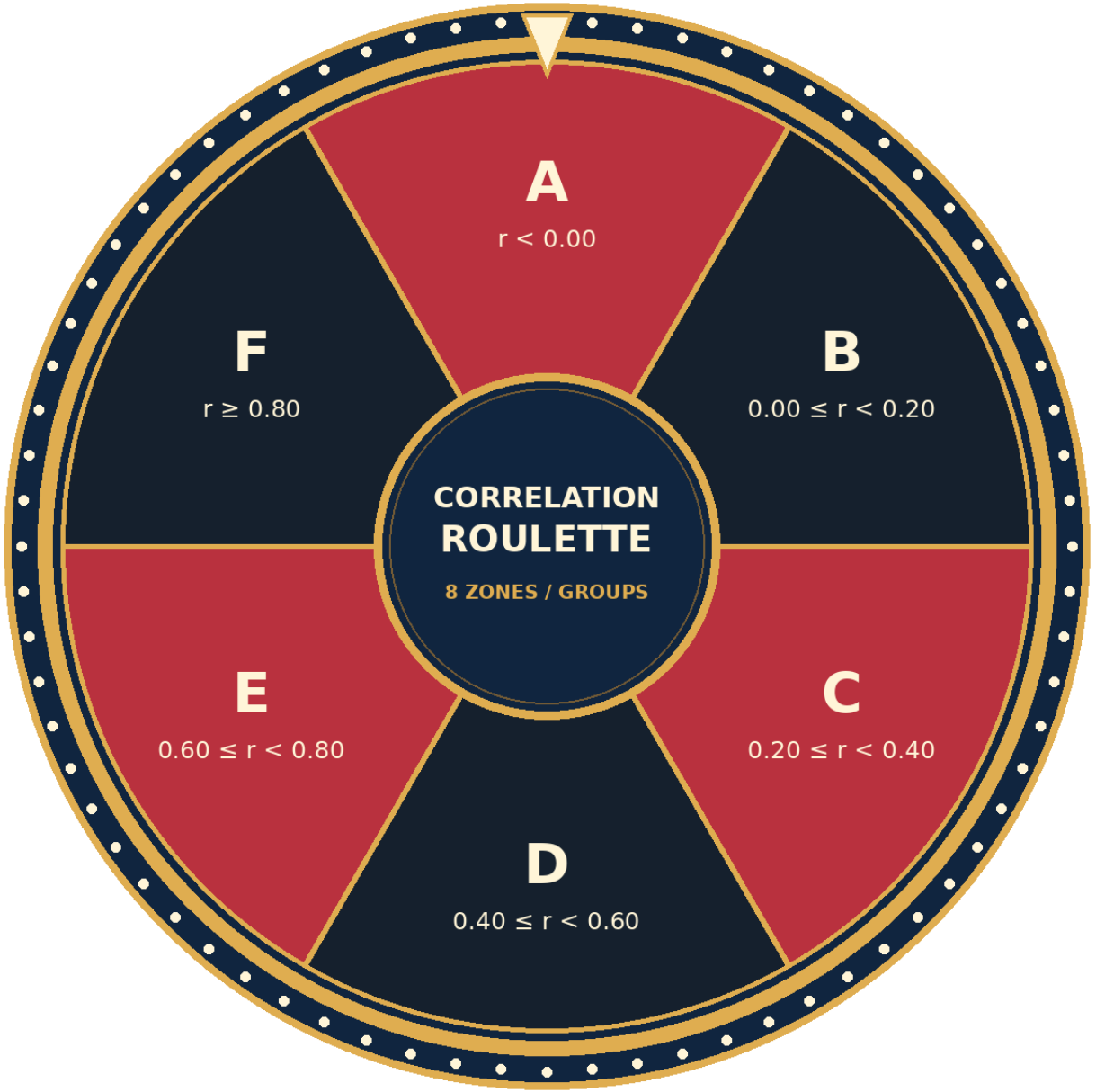
DAY 1 AT THE SPATIAL CASINO

Let's try out the correlation roulette!

ACTIVITY

Game Rules:

- Form six teams. Each team starts with **50** points.
- Each round, predict the correlation coefficient produced by combining the same Iowa counties into **8 zones/groups**.
- Choose one correlation range:
  - A: $r < 0.00$
  - B: $0.00 \leq r < 0.20$
  - C: $0.20 \leq r < 0.40$
  - D: $0.40 \leq r < 0.60$
  - E: $0.60 \leq r < 0.80$
  - F: $r \geq 0.80$



ACTIVITY

Game Rules:

- Bet **1–15 whole-number points**, without exceeding your current balance.
- Give the TA your team number, selected range, and wager. The TA places your sticky note on the board.
- Once all bets are submitted, bets are locked!
- After bets are locked, the class chooses an algorithm by rolling a dice
  - **Zoning (1-3)**
  - **Grouping (4-6)**

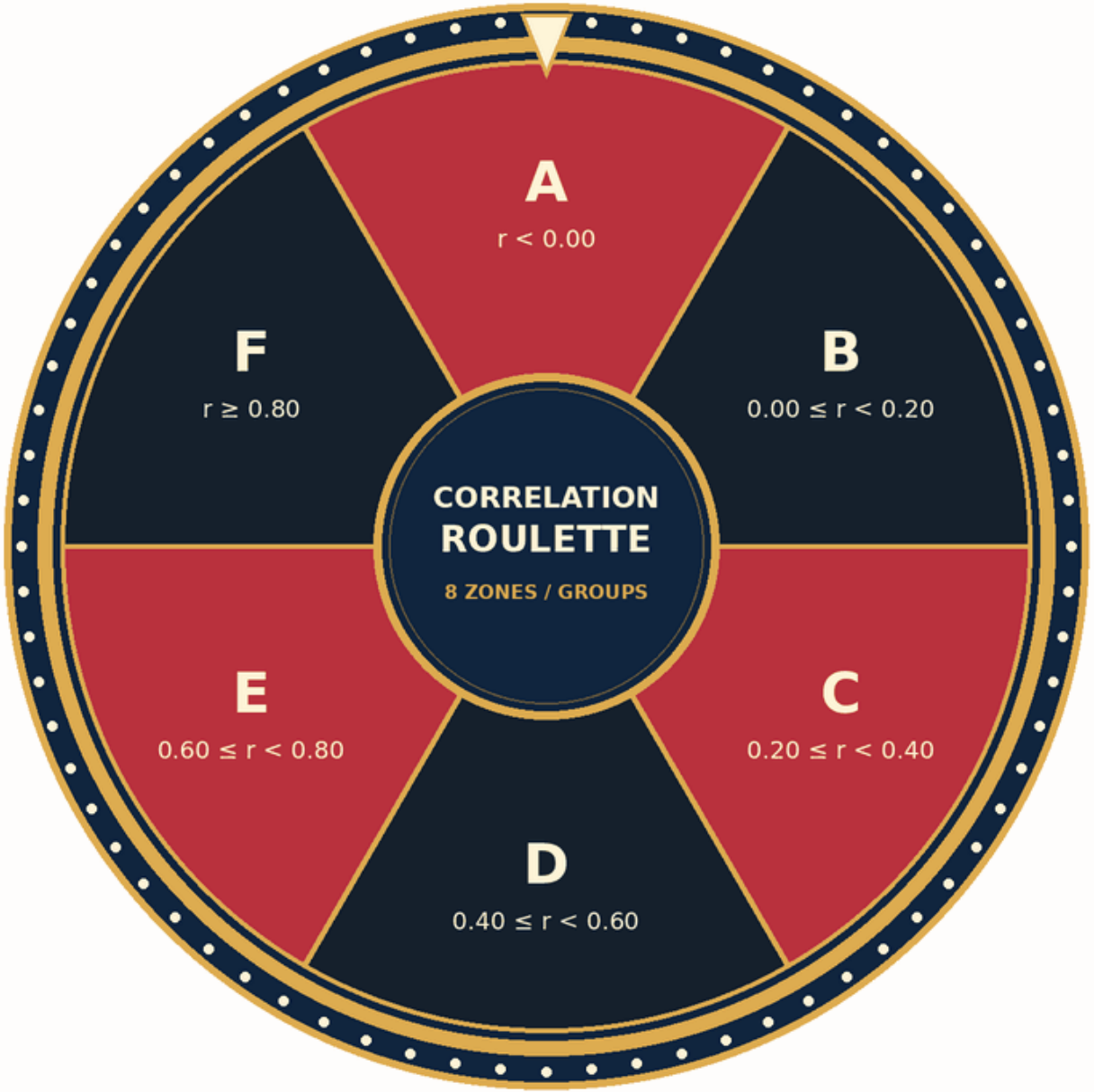


ACTIVITY

Game Rules:

- The instructor runs the selected algorithm once and reveals the resulting r and identifies the winning range.
- **Correct prediction:** gain the number of points wagered.
- **Incorrect prediction:** lose the number of points wagered.

Example: Bet 10 points on D. If r = 0.53, gain 10 points; if r = 0.73, lose 10 points.



DEBRIEF

- What distribution of correlation coefficients do you think repeated runs would produce?
- Would zoning and grouping produce similar or different distributions? Why?
- What features of the underlying data-generating process might make extreme correlations possible?
- If the number of zones stays fixed at 8, why do we still see different values of r?



DAY 2 AT THE SPATIAL CASINO

Let's win the MAUP Jackpot!

ACTIVITY

Game Rules:

- Your team now controls the experiment using the R simulation code.
- Your challenge: produce the highest positive or lowest / most negative correlation, according to the target announced by the instructor.
- Unlike roulette, you may now change the number of zones/groups, choose zoning or grouping
- **Before you start anything,** your dealer in training explains:
  - Where to change each setting.
  - What each setting changes about the experiment.
  - Where to find the resulting correlation.
  - **Do not run the code yet!** Make sure everyone understands the choices before starting.

ACTIVITY

STEPS:

CHOSE SETTINGS (30s)	Teams discuss what number of zones and algorithm settings they think will produce an extreme result.
RECORD SETTINGS (20s)	Teams must write down their chosen settings before they run the simulation.
RUN SIMULATIONS (20s)	Teams run as many simulations as they can using the recorded settings.
REPORT	Teams report the number of simulations they ran and their most extreme outcome.

DEBRIEF

- Which settings produced the most extreme outcomes?
- Did the same settings work for every team, or did randomness still matter?
- Did changing the number of zones make it easier to produce extreme correlations?
- How did zoning and grouping differ?
- What does this reveal about the difference between the **aggregation problem** and the **scale problem**?

https://docs.google.com/spreadsheets/d/1TOLeJwr3d4y80JgPQs2E2xOON-6rtQf92UFIAn6XZGQ/edit?usp=sharing

TAKEAWAYS

- Researcher decision impact the outcome of geographic analyses, but not all decisions have the same impact.
- You can now differentiate between the **scale problem** and the **aggregation problem** and consider both in your own practices.

TUTORIAL

In this tutorial, we will reproduce and extend Openshaw (1970)'s first experiment on the Modifiable Areal Unit Problem (MAUP).

- § 3: Conduct exploratory data analysis to examine the distributions and spatial patterns of the two variables of interest.
- § 4: Investigate the scale problem by applying the zoning and grouping algorithm to generate alternative geographies and compute the correlation coefficients.
- § 5, Investigate the aggregation problem by applying the zoning and grouping algorithm to generate multiple geographies and computing the correlation coefficients for each.
- § 6: Combine the scale and aggregation problems to generate 1,000 geographies of the same number of zones and repeat this process for different numbers of zones.